

1. Key Findings

After looking at all the plots, a few patterns stood out pretty clearly. First, students who study more hours per week usually have a higher Current_GPA. The regression line for study hours vs GPA goes up, and most low-study students sit in the lower GPAs, which feels kinda expected but still good to confirm with data.

Attendance also matters a lot. Students with higher Attendance_Rate usually end up with higher Final_Average scores. When attendance drops, grades get way more unpredictable and mostly lower. Another strong pattern came from comparing Previous_GPA with Current_GPA — students who did well before are almost always the ones doing well now too.

Sleep seems to play a role too. Students who sleep way too little usually have weaker GPAs, while those who get a normal amount of sleep (not too much, not too little) seem to do better. It's not a perfect straight line, but the trend is there.

Different majors also show different GPA levels. Some majors have higher medians while others are lower. Same for year levels — freshmen seem to struggle a bit more, which kinda makes sense since they're still getting used to everything. And students who work a lot of part-time hours often fall into the lower GPA range, likely because they just have less time/energy.

2. Pattern Interpretation

The main idea here is that good habits tend to show up again and again. High study hours, good attendance, and a strong previous GPA almost always line up with better outcomes. One interesting thing was the “sweet spot” for sleep — too little sleep clearly hurts performance, but too much doesn't magically boost GPA either.

Freshmen showing lower results wasn't surprising, but it's still important. Some majors being lower might mean they're harder or maybe students need more support there. Students who work many hours appear to struggle more, which might be because they simply don't have enough time left.

3. Actionable Recommendations

For Students

- Try to study steadily each week instead of cramming.
- Don't skip classes — attendance really matters.
- Get enough sleep (seriously).
- If you work many hours, try organizing your schedule so studying doesn't get pushed out.

For the University

- Give first-years more guidance or workshops.
- Provide tutoring for majors with lower GPA averages.
- Support students who work a lot, maybe with flexible deadlines or extra advising.
- Continue offering scholarships, since scholarship students seem to do a bit better.

For At-Risk Groups

- Watch for low-attendance students early.
 - Offer mentoring for students with low Previous_GPA.
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4. Limitations

The dataset doesn't include things like mental health, stress, motivation, or teaching quality, which all matter a lot. Also, these graphs show correlation, not cause. Just because studying more is linked to higher GPA doesn't automatically mean one causes the other. More info about lifestyle or personal challenges would make the analysis stronger.